

Ernesto E. Vargas-Parra

Postdoctoral Fellow
University of Toronto
Department of Ecology and Evolutionary Biology
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Education

American Museum of Natural History, Richard Gilder Graduate School 2019-2024

Ph.D. in Comparative Biology

Dissertation: Modularity at Different Levels in Trilobites

Advisor: Dr. Melanie J. Hopkins

University of California, Riverside 2017-2019

M.S. in Geological Sciences

Thesis: Exploring Variation in Late Cambrian Trilobite *Dikelocephalus* Pygidia Using Landmark-Based Geometric Morphometrics

Advisor: Dr. Nigel C. Hughes

University of Chicago 2013-2017

B.A. in Geophysical Sciences, with honors

Honors Thesis: Discrimination of Intraspecific Variation from Interspecific Disparity: A Case Study Using the *Olenellus fowleri* Species Complex (Trilobita, Early Cambrian)

Advisor: Dr. Mark Webster

Professional Positions

Postdoctoral Fellow 2026-Present

University of Toronto

Department of Ecology and Evolutionary Biology

Advisor: Dr. C. Tomomi Parins-Fukuchi

Chancellor's Postdoctoral Fellow 2024-2026

University of California, Riverside

Department of Earth and Planetary Sciences

Advisor: Dr. Nigel C. Hughes

Research Associate 2024-Present

American Museum of Natural History

Division of Paleontology

Publications

Vargas-Parra, E. E., Surprenant, R. L., Leung, R., Taylor, N., García-Bellido, D. C. and Droser, M. L. *In review*. Decoupling of formation-level occurrences of *Rusophycus* and trilobite body fossils through time: implications for secular change in sediment rheology. *Palaeontology*.

- Srivastava-Losey, S., **Vargas-Parra, E. E.** and Hughes, N. C. *In review*. Phylogenetic analysis of the late Cambrian Laurentian trilobite family Dikelocephalidae Miller, 1889. *Journal of Paleontology*.
- Hopkins, M. J., Nikolic M. C., Holmes, J. D., Monti, D. S., **Vargas-Parra, E. E.**, Bicknell, R. D. C., Edgecombe, G. D., Jordan-Burmeister, K., Paterson, J. R. and Srivastava-Losey, S. 2026. Building phenotypic character matrices for phylogenetic inference: exploration of 35 years of practice. *Biological Reviews*, 101, 2379–2407.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2026. Modular reorganization of the trilobite *Lonchodomas chaziensis* demonstrates mosaic patterns of phenotypic change underlie metamorphosis. *Evolution*, qpag018. *Editor's Choice article for April issue of Evolution*.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2025. The modular organization of the trilobite head in *Ceraurus pleurexanthemus*. *Palaeontology*, 68, e70013.
- Westrop, S. R., Amati, L. M., and **Vargas-Parra, E. E.** 2025. Upper Ordovician (Sandbian–Katian) species of the trilobite *Calyptaulax* Cooper, 1930 (Pterygometopidae) from the central United States and Canada. *Australasian Palaeontological Memoirs*, 57, 221–250.
- Bicknell, R. D. C., **Vargas-Parra, E. E.**, Landman, N. H., and Pärnaste, H. 2024. Evidence for cryptic moulting behavior in the trilobite *Toxochasmops vormsiensis* from the Upper Ordovician Katian Kõrgessaare Formation, Estonia. *The Science of Nature*, 111, 22.
- Fortey, R. A., **Vargas-Parra, E. E.** and Droser, M. L. 2024. Trilobites from the Al Rose Formation (Lower Ordovician, Inyo Mountains, California) – faunas marginal to the Great Basin. *Journal of Paleontology*, 98, 644–657.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2022. Modularity in the trilobite head consistent with the hypothesized segmental origin of the eyes. *Evolution & Development*, 24, 177–188. *Editor's Choice article for November issue of Evolution & Development*.

Teaching and Mentorship

Instructor of Record:

University of California, Riverside
GEO 003/BIO 010 Headlines in the History of Life (Summer 2026)

Teaching Assistant:

Columbia University
EESC UN2300 Earth's Environmental Systems: Life Systems (Spring 2021)
EESC UN1001/1401 Dinosaurs and the History of Life (Fall 2021)

Mentoring:

Marisol Ibarra Zea, UCR, master's student mentee	2024-Present
Kian McElfresh, UCR, undergraduate research assistant	2025-Present

Fellowships, Awards, and Grants

Editor's Choice Award for April issue of <i>Evolution</i>	2026
Chancellor's Postdoctoral Fellowship (University of California, Riverside)	2024
President's Student Research Award (Paleontological Society)	2023
Editor's Choice Award for November issue of <i>Evolution & Development</i>	2022
Lerner-Gray Marine Research Grant (American Museum of Natural History)	2022
Geological Society of America 2021 Conference Grant (Paleontological Society)	2021
Graduate Trainee Fellow (American Museum of Natural History)	2019
Dean's Distinguished Fellowship Award (University of California, Riverside)	2017
Gates Millennium Scholar Graduate Fellowship (Bill & Melinda Gates Foundation)	2017
UCISTEM Summer Research Grant (University of Chicago)	2016
Gates Millennium Scholar (Bill & Melinda Gates Foundation)	2013
Odyssey Scholar (University of Chicago)	2013

Invited Talks

Vargas-Parra, E. E. 2026. Metamorphosis, mosaic change, and modular reorganization of the trilobite *Lonchodomas chaziensis*. Raymond M. Alf Museum of Paleontology.

Vargas-Parra, E. E. 2026. Macroevolutionary consequences of extinction on trilobite modularity. University of Toronto.

Vargas-Parra, E. E. 2026. Evolutionary modularity of a trilobite family over the Ordovician radiation and Late Ordovician mass extinction. University of Birmingham.

Vargas-Parra, E. E. 2025. Getting the Most Out of the Fossil Record. University of California, Riverside.

Vargas-Parra, E. E. 2023. Trilobite Modularity Over the Ordovician Radiation. Brooklyn College, City University of New York.

Conference Abstracts

Vargas-Parra, E. E. and Hopkins, M. J. 2026. Evolutionary modularity of a trilobite family over the Ordovician radiation and Late Ordovician mass extinction. 7th International Trilobite Meeting, San Emiliano, Spain.

Hughes, N. C., Srivastava-Losey, S. and **Vargas-Parra, E. E.** 2026. Phylogenetic analysis of the late Cambrian Laurentian trilobite family Dikelocephalidae Miller 1889 and evolutionary patterns among some of its members. 7th International Trilobite Meeting, San Emiliano, Spain.

Vargas-Parra, E. E., Srivastava, S., Hughes, N. C. and Runkel, A. C. 2025. Microevolutionary trajectories in the late Cambrian complex trilobite species *Dikelocephalus minnesotensis*. Geological Society of America Annual Meeting 2025, San Antonio, TX, USA.

Srivastava, S., **Vargas-Parra, E. E.**, Hughes, N. C. and Runkel, A. C. 2025. Reign of trilobites in Upper Mississippi Valley: Phylogeny of the late Cambrian trilobite family Dikelocephalidae Miller, 1889. Geological Society of America Annual Meeting 2025, San Antonio, TX, USA.

Hopkins, M. J., Nikolic M. C., Holmes, J. D., Monti, D. S., **Vargas-Parra, E. E.**, Bicknell, R. D. C., Edgecombe, G. D., Jordan-Burmeister, K., Paterson, J. R. and Srivastava-Losey, S. 2025. Building phenotypic character matrices for phylogenetic inference: exploration of 35 years of practice. Geological Society of America Annual Meeting 2025, San Antonio, TX, USA.

- Hughes, N. C., **Vargas-Parra, E. E.**, Srivastava, S. and Runkel, A. C. 2024. Evolutionary Tempo and Mode in Cambrian Upper Mississippi Valley Dikelocephalid Trilobites I: *Dikelocephalus*. Geological Society of America Annual Meeting 2024, Anaheim, CA, USA.
- Srivastava, S., **Vargas-Parra, E. E.**, Hughes, N. C. and Runkel, A. C. 2024. Evolutionary Tempo and Mode in Cambrian Upper Mississippi Valley Dikelocephalid Trilobites II: *Osceolia* and *Walcottaspis*. Geological Society of America Annual Meeting 2024, Anaheim, CA, USA.
- Leung, R., Surprenant, R. L., **Vargas-Parra, E. E.**, Taylor, N., García-Bellido, D. C. and Droser, M. L. The inverse relationship between the stratigraphic distribution of *Rusophycus* and bioturbation through time: Implications for secular change in sediment rheology. Geological Society of America Annual Meeting 2024, Anaheim, CA, USA.
- Hughes, N. C., Srivastava, S., **Vargas-Parra, E. E.** and Runkel, A. C. 2024. Evolutionary Tempo and Mode in Stratigraphic Context: Dikelocephalid Trilobite Evolution in the Upper Mississippi Valley Cambrian. 12th North American Paleontological Convention 2024, Ann Arbor, MI, USA.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2024. Phylogeny and evolutionary modularity of a trilobite family over the Ordovician Radiation. 12th North American Paleontological Convention 2024, Ann Arbor, MI, USA.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2023. Phylogeny and evolutionary modularity of a trilobite family over the Ordovician Radiation. Geological Society of America Annual Meeting 2023, Pittsburgh, PA, USA.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2023. Phylogeny and evolutionary modularity of a trilobite family over the Ordovician Radiation. 14th International Symposium on the Ordovician System, Tallinn, Estonia.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2022. The modular organization of the trilobite head. Geological Society of America Annual Meeting 2022, Denver, CO, USA.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2022. The modular organization of the trilobite head. 6th International Palaeontological Congress, Khon Kaen, Thailand.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2021. Ontogenetic modularity and metamorphosis in an Ordovician trilobite. Geological Society of America Annual Meeting 2021, Portland, OR, USA.
- Vargas-Parra, E. E.** and Hopkins, M. J. 2020. Testing hypotheses of trilobite head modularity with emphasis on the eyes. 64th Palaeontological Association Annual Meeting, Oxford, UK.
- Vargas-Parra, E. E.** and Hughes, N. C. 2019. Exploring Variation in Late Cambrian Trilobite *Dikelocephalus* Pygidia Using Landmark-Based Geometric Morphometrics. 11th North American Paleontological Convention 2019, Riverside, CA, USA.
- Hernandez, A. V., Bozer, R. A., Dettman, M. C., Drummer, K. Y., Rubin, S. E., **Vargas-Parra, E. E.**, O'Leary, R., Landman, N. H., Hopkins, M. J. and Hussaini, B. M. 2017. Curation of the Royal H. Mapes invertebrate paleontology collection at the American Museum of Natural History: strategies for digitization and outreach. Geological Society of America Annual Meeting 2017, Seattle, WA, USA.

Peer Reviewer for

Journal of Paleontology; Historical Biology; Royal Society Open Science; Palaeogeography, Palaeoclimatology, Palaeoecology; Swiss Journal of Palaeontology; Papers in Palaeontology

Museum Collection Study Visits

Peabody Museum, Yale University (USA); Museum of Comparative Zoology, Harvard University (USA); Smithsonian Natural History Museum (USA); Field Museum of Natural History (USA); Sedgwick Museum, University of Cambridge (UK); University of Bristol (UK); Royal Scottish Museum (UK); Natural History Museum (UK); Lapworth Museum of Geology, University of Birmingham (UK); British Geological Survey, Keyworth (UK); British Geological Survey, Edinburgh (UK); Natural History Museum of Estonia (Estonia); Tallinn University of Technology (Estonia); University of Tartu Natural History Museum (Estonia); Natural History Museum at the University of Oslo (Norway); Swedish Museum of Natural History (Sweden); National Museum (Czech Republic); Czech Geological Survey (Czech Republic); Royal Belgian Institute of Natural Sciences (Belgium); Museum für Naturkunde Berlin (Germany)

Outreach and Professional Service

- Earth-sciences Reciprocal Learning Year Initiative (EARLY) 2019-2024
- Traveled with high school teachers to do fieldwork in Upstate New York to find fossils and collect bulk rock at the Penn Dixie Fossil Park & Nature Reserve
 - Visited NYC middle and high school classrooms to discuss and help identify fossils as students broke up rock to find fossils firsthand
- Adventures in Science summer camp, American Museum of Natural History 2022-2023
- Presented to students about the tools I use as a paleontologist
 - Demonstrated such tools including rock hammers and 3D surface scanners
- #LearnWithMe: Take a Trilobite Tour, American Museum of Natural History Summer 2020
- Virtual tour of behind-the-scenes fossil collections of the AMNH
 - www.youtube.com/watch?v=4x4xI9KupaY
- Geoscience Development (GEODE) Program, University of California Riverside Spring 2019
- Provided science instruction on high school field trips to Death Valley, Nevada
- Geological Education Outreach Program (GEOP), UC Riverside 2018-2019
- Elementary school presentations on geoscience
 - Community climate fairs & education events
- Student Organizer, 11th North American Paleontological Convention Summer 2019
- Helped organize and ran various tasks before and during the conference
- Campus-Based Leader, Gates Millennium Scholars Program 2014-2016
- Served as a campus liaison connecting the University of Chicago with the Gates Program
 - Directed Gates Scholars to student support services and other resources on campus
- Community Member, Sam Noble Oklahoma Museum of Natural History Spring 2014
- Represented the museum by sharing personal involvement in museum programs and events
 - Represented the museum at the White House by accepting the National Medal for Museum and Library Service awarded by the Institute of Museum and Library Services (IMLS)